Lasso Trading

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1 Introduction

The volume and granularity of data available worldwide is growing at an unprecedented pace. In 2024 alone, the world generated approximately 402 million terabytes of data per day, up from 43 million just a decade earlier (Wright, 2025). Financial markets are no exception to this trend. Whereas investors once had access to only a handful of data series, today they can draw on a vast array of signals, ranging from traditional balance-sheet information to textual data, satellite imagery, and web traffic. For most of these new variables, there is no clear theoretical prior about their relationship, if any, to future stock returns. Nevertheless, such data are in high demand with investors. The reason is that investors increasingly rely on machine learning and artificial intelligence (AI) to find useful patterns in large datasets.[1] These methods require no theorytical guidance about which variables should matter; instead, they let the data speak for themselves.

This behavior is not well captured by existing asset pricing models with learning. In these models, agents typically are assumed to employ self-referential or correctly specified forecasting rules, or to work with severely under-parameterized perceived laws of motion. Agents track a small number of parameters and update them recursively using variants of least squares or constant-gain learning. These frameworks are analytically convenient, but they do not describe investors who suspect that some among thousands of potential signals may forecast returns and who use flexible statistical procedures to discover such relationships without an explicit structural model in mind.

We develop a model that embeds this form of statistical learning into a standard asset pricing environment. Under rational expectations, returns in our economy are not predictable. Agents, however, suspect that returns may be conditionally predictable using a large set of potential predictors. To exploit this perceived opportunity, they bet on sparsity and apply a machine learning method, least absolute shrinkage and selection operator (LASSO), to select a sparse subset of variables from the high-dimensional predictor space and to construct forecasting rules (Tibshirani, 1996).

Our main result is that the very act of searching for predictability with such statistical learning methods can generate short-lived pockets of return predictability. In equilibrium, these pockets are sparse and evolve over time, echoing recent empirical evidence on time-varying short-term predictability in financial markets (Chinco, Clark-Joseph and Ye, 2019; Farmer, Schmidt and Timmermann, 2023). In essence, the process of learning in high-dimensional data both hunts for and endogenously creates the temporary predictability

[1]The Invesco Global Systematic Investing Study 2024 reports that more than half of surveyed investors incorporate AI, using tools ranging from pattern recognition to portfolio optimization (Invesco, 2024).

it seeks to exploit, providing a new perspective on how modern data-driven investment practices interact with asset prices.

We provide what is, to our knowledge[2], the first theoretical characterization of LASSO-based learning in an equilibrium asset-pricing environment. This allows us to link a literature that has relied on low-dimensional, stylized learning rules with the empirical tradition in which researchers and practitioners routinely estimate high-dimensional predictive models. Endowing agents with a realistic learning rule is not only conceptually appealing; it also enables a tight connection between theory and data. On the theoretical side, we show that LASSO learning generates pockets of predictability even when returns are fundamentally unpredictable. We analytically characterize the probability with which such pockets arise, providing a precise mechanism that rationalizes the sparse and episodic nature of return predictability documented in recent literature. This framework delivers strong predictions regarding when and how predictability emerges in finite samples, which we test using a wide set of assets and a rich vocabulary of potential predictors constructed from textual "topics" (Bybee et al., 2021).

The Model Here comes another description of the model. This time in more detail, describing baseline, learning mechanism and equilibrium.

Results Here comes a short summary of our most important results.

Related Literature There is a long-standing tradition of asset-pricing models incorporating learning mechanisms, to which our model is closely related. The premise of this literature is that agents possess incomplete knowledge of the data-generating process for market outcomes and therefore attempt to infer it from observed data. Typically, learning is introduced to for prominent asset-pricing anomalies, including excess volatility, return predictability, and the time-varying price–dividend ratio, that are hard to reconcile with agents that know how the economy works. Within this tradition, models vary in their assumptions about what agents know about the true model, the learning technology they use, and what specific variable they aim to learn.

The earliest contributions to this literature featured perfectly rational agents that learn about an unknown parameter of a known model using least squares estimation (Marcet and Sargent, 1989). Timmermann (1993, 1996) shows that even in this simple setting learning can generate excess volatility and return predictability. The learning problem considered here is low-dimensional and converges to the rational expectations equilibrium (REE) as agents see more data. Convergence to the REE is not guaranteed when agents are not fully rational. Honkapohja and Mitra (2003) show that if agents have limited memory and can only use a finite sample of past observations, the model does not necessarily converge to the REE. The asset pricing implications of learning models become richer when agents are assumed to not only be uncertain about model paramters but also about whcih model is the correct one. Barberis, Shleifer and Vishny (1998) study agents that switch between

[2] Georges and Pereira (2021) study by means of simulation various types of regularization methods, including LASSO in a model in which forecasting rules are auto-regressive including higher-rder polynomials of past returns to allow for non-linearities. Compared to them we provide analytical results and focus on the use of LASSO in linear regression on high-dimensional data.

two different earnings forecasting rules. Hong, Stein and Yu (2007); Branch and Evans (2010) study models in which agents estimate a univariate perceived law of motion (PLM) for returns, while the true model is multivariate.

More recently, Adam and Marcet (2011) pointed out that the standard learning framework implicitly assumes that agents know how fundamentals map to market outcomes. However, given that most economists cannot agree on a definitive asset pricing model, this assumption is questionable. Adam, Marcet and Nicolini (2016) pick up on this argument and propose a model which integrates agents are ”internally rational” into a standard Lucas (1978) asset pricing framework. These agents have subjective beliefs about variables beyond their control and maximize utility based on these beliefs. Crucially, these agents do not know how fundamentals map to prices; instead, they learn this mapping from observed prices. This learning mechanism creates a feedback loop between beliefs and prices, enabling the model to match various asset-pricing moments.

We use Adam, Marcet and Nicolini (2016) as the starting points for our theory of LASSO trading. Doing so has two advantages: First, we aim to model the impact of agents using machine learning techniques to form expectations on market outcomes. Such a feedback loop emerges in Adam, Marcet and Nicolini (2016) already under a very general learning rule which we replace with a microfounded LASSO learning mechanism. Second, their framework assumes that agents understand the dividend process, yet are boundedly rational regarding returns. This enables us to distinguish the impact of agents employing misspecified forecasting rules for returns from mispecification regarding dividends.

Our model is motivated by the empirical evidence that return predictability is an unstable phenomenon. While traditional variables such as consumption growth, dividend yields, and short-term interest rates forecast returns in certain periods, their predictive power vanishes in others (Ang and Bekaert, 2007; Goyal and Welch, 2003; Lettau and Ludvigson, 2001). More recent evidence points to sparsity as the second key feature of return predictability.

More recenlty, researchers have documented similar patterns at high frequencies. Chinco, Clark-Joseph and Ye (2019) show that return predictability at high frequencies is unstable and concentrated in a small and shifting set of predictors. Using LASSO regressions on a large set of candidate predictors at a high frequency, they found that predictive relationships emerge briefly and cannot be captured by a fixed forecasting model. The frequency of our model is somehow in the middle as we consider daily return series. These findings suggest that return predictability can be explained by models in which agents search for and exploit fleeting patterns in large datasets, as we do in this paper.

2 The model

We study a standard Lucas (1978) asset pricing model in the flavour of Adam, Marcet and Nicolini (2016). We consider an economy populated by a continuum of agents of mass one who trade a single risky asset. The asset pays a stochastic dividend D_t at each discrete time period $t = 0, 1, 2, \ldots$. The dividend and consumption processes are driven by multivariate normally distributed shocks

$$\begin{pmatrix} \log \varepsilon_t^d \\ \log \varepsilon_t^c \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} -s_d^2/2 \\ -s_c^2/2 \end{pmatrix}, \begin{pmatrix} s_d^2 & \rho s_d s_c \\ \rho s_d s_c & s_c^2 \end{pmatrix} \right), \tag{1}$$

The dividend process follows

$$\frac{D_t}{D_{t-1}} = a\varepsilon_t^d, \tag{2}$$

where $a \geq 1$ is a constant growth factor, which implies that

$$\mathbb{E}_t \left[D_{t+1} \right] = aD_t. \tag{3}$$

Similarly for the aggregate consumption process we postulate that

$$\frac{C_t}{C_{t-1}} = a\varepsilon_t^c. \tag{4}$$

The equilibrium price of the asset is determined by the standard no-arbitrage condition which requires that the price at time t equals the discounted expected value of next period's price plus dividend, that is

$$P_t = \delta \mathbb{E}_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} P_{t+1} + D_{t+1} \right], \tag{5}$$

where δ is the discount factor and $\mathbb{E}_t[\cdot]$ is the expectation operator conditional on information available at time t.

Under rational expectations, the equilibrium price can be solved as

$$P_t = \frac{\delta a^{(1-\gamma)} \rho_e}{1 - a^{(1-\gamma)} \rho_e} D_t, \tag{6}$$

where

$$\rho_e = \mathbb{E}_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} \frac{D_{t+1}}{D_t} \right] = e^{\gamma(1+\gamma)s_c^2/2} e^{-\gamma \rho s_c s_d}. \tag{7}$$

This implies a constant price-dividend ratio and an ex-dividend gross return which is given by

$$R_{t+1} = \frac{P_{t+1}}{P_t} = \frac{D_{t+1}}{D_t} = a\varepsilon_{t+1}^d. \tag{8}$$

Taking logs we obtain log-returns

$$r_{t+1} = \log(R_{t+1}) = \log(a) + \log(\varepsilon_{t+1}^d) \sim \mathcal{N}(\log(a) - s_d^2/2, s_d^2). \tag{9}$$

Without loss of generality in what follows we set $\log(a) = s_d^2/2$. This implies that under rational expectations log-returns have mean zero and there is no time-t information that can help forecasting returns.

Instead of assuming that agents know the true data generating process, we follow the adaptive learning literature and assume that agents are boundedly rational and form their expectations using econometric techniques [citations here]. Equation (5) shows that agents have to forecast two quantities, the risk-adjusted gross return and gross dividend growth. We follow Adam, Marcet and Nicolini (2016) in making the following assumption.

Assumption 1 *Agents know the process for dividends and consumption and are boundedly rational only with respect to (gross) returns.*

In this way we can isolate the effects of learning about returns on asset prices. The usual approach to learning is to endow agents with a Perceived Law of Motion (PLM) which is of the same functional form as the Actual Law of Motion (ALM) but with unknown parameters that agents estimate using available data. In this paper we depart from the standard approach and assume that agents use linear models and try to forecast log-returns on some set of predictors available at time t.

2.1 Univariate PLM and ALM

To fix ideas we start by considering the univariate case in which agents try to predict log-returns using a single predictor x_t.

Specifically we assume that agents have a linear and stationary PLM for the asset return

$$r_{t+1} = x_t\beta + u_t, \tag{10}$$

with $x_t \sim \mathcal{N}(0, \sigma_x^2)$ being a signal available for the agent at time t which they suspect might have predictive power on log-returns.

Then the agents' PLM about log-returns implies the following PLM for gross returns

$$R_{t+1} = e^{x_t\beta + u_t}. \tag{11}$$

This implies that gross expected returns follow a log-normal distribution and their conditional forecast is

$$\mathbb{E}_t(R_{t+1}) = \mathbb{E}_t\left(\frac{P_{t+1}}{P_t}\right) = e^{x_t\beta + \sigma_u^2/2} = \phi e^{x_t\beta}, \tag{12}$$

by letting $\phi = e^{\sigma_u^2/2}$.

Assumption 2 *Within their perceived law of motion, agents believe that the return process and consumption growth are independent.*

Assumption 2 allows us to derive risk-adjusted expected gross returns as

$$\mathbb{E}_t\left[\left(\frac{C_{t+1}}{C_t}\right)^{-\gamma}\frac{P_{t+1}}{P_t}\right] = \mathbb{E}_t\left[\left(\frac{C_{t+1}}{C_t}\right)^{-\gamma}\right]\mathbb{E}_t\left(\frac{P_{t+1}}{P_t}\right) = a^{-\gamma}\phi e^{x_t\beta}. \tag{13}$$

Plugging this into (5) yields

$$P_t = \frac{\delta a^{(1-\gamma)} \rho_e}{1 - \delta a^{-\gamma} \phi e^{x_t \beta}} D_t. \tag{14}$$

Now divide by P_{t-1} to obtain actual gross returns

$$R_t = \frac{P_t}{P_{t-1}} = \frac{1 - \delta a^{-\gamma} \phi e^{x_{t-1} \beta}}{1 - \delta a^{-\gamma} \phi e^{x_t \beta}} \frac{D_t}{D_{t-1}}, \tag{15}$$

and taking logs and moving one step forward we obtain the ALM

$$r_{t+1} = \log(\varepsilon_{t+1}) + \log(1 - \kappa e^{x_t \beta}) - \log(1 - \kappa e^{x_{t+1} \beta}), \tag{16}$$

where $\kappa := \delta a^{-\gamma} \phi$.

2.2 Global well-definedness and bounded returns

Expression (16) is only well-defined if

$$1 - \kappa e^{x_t \beta} > 0 \quad \text{and} \quad 1 - \kappa e^{x_{t+1} \beta} > 0$$

hold with probability one, which imposes state-dependent restrictions on (x_t, β). To avoid these domain issues and to guarantee global existence of a stochastic equilibrium, we adopt a bounded specification for the conditional mean of returns.

We modify the perceived conditional expectation of gross returns as

$$\mathbb{E}\left(R_{t+1} \mid \mathcal{F}_t\right) = \phi \exp\Big(w \tanh(x_t \beta / w)\Big), \tag{17}$$

where

$$w = -\log(\kappa) - \varepsilon, \qquad \varepsilon > 0.$$

Since $\tanh(\cdot) \in (-1, 1)$, this implies

$$w \tanh(x_t \beta / w) \in (-w, w) \quad \Rightarrow \quad e^{w \tanh(x_t \beta / w)} \in (e^{-w}, e^{w}).$$

By construction $w < -\log(\kappa)$, so

$$e^{w} < e^{-\log(\kappa)} = \kappa^{-1},$$

and therefore

$$1 - \kappa e^{w \tanh(x_t \beta / w)} > 0 \quad \text{for all } x_t, \beta.$$

The corresponding risk-adjusted expectations and price-dividend ratio are defined exactly as above, with $x_t \beta$ replaced by $w \tanh(x_t \beta / w)$. The implied (modified) ALM is

$$r_{t+1} = \log(\varepsilon_{t+1}) + \log\Big(1 - \kappa e^{w \tanh(x_t \beta / w)}\Big) - \log\Big(1 - \kappa e^{w \tanh(x_{t+1} \beta / w)}\Big). \tag{18}$$

For later use, define the function

$$g_\beta(x) := \log\Big(1 - \kappa\, e^{w \tanh(x\beta/w)}\Big). \tag{19}$$

This function is globally well-defined and smooth in x and β.

Remark 1 *For small σ_x^2 and moderate values of κ, the original ALM (16) is well-defined with high probability. The bounded specification (18) should then be interpreted as a globally valid extension that coincides with the baseline model in the empirically relevant region of the state space but eliminates corner cases where the log is undefined.*

Clearly this ALM is non-linear, which implies that the linear PLM used by agents is misspecified. In these cases it is unlikely that the learning procedure will converge to the REE and the literature has instead focused on convergence to a Stochastic Consistent Expectations Equilibrium (SCEE) [citations here]. In our case we look for an equilibirum which is cross-moment consistent in the sense that, despite their misspecification, there is a statistical self-consistency between the empirical relationship they perceive and the one which is implied by the ALM. Analogously to the SCEE case, we define a Cross-Moment Consistent Equilibrium (CMCE) as follows.

Definition 1 (Cross-Moment Consistent Equilibrium) *Let $\mathcal{R}$ denote a regression operator mapping a joint distribution over (r_{t+1}, x_t) into a parameter vector.*[3]

A pair (μ, β) is a Cross-Moment Consistent Equilibrium (CMCE) *associated with $\mathcal{R}$ if:*

1. *μ is a non-degenerate invariant probability measure for the stochastic process*

$$r_{t+1} = \log(\varepsilon_{t+1}) + g_\beta(x_t) - g_\beta(x_{t+1}).$$

2. *The parameter β is fixed by the regression operator applied to the invariant measure, that is,*

$$\beta = \mathcal{R}(\mu),$$

where $\mathcal{R}(\mu)$ denotes the population regression coefficient obtained by applying $\mathcal{R}$ to the joint distribution of (r_{t+1}, x_t) induced by μ.

Intuitively, while a REE imposes consistency on the *entire* distributions implied by the ALM and PLM, a CMCE only requires consistency on certain moments of the two distributions, and in particular of those moments that are relevant for the agent's forecasting problem.

We now turn to characterizing existence and stability of CMCEs under different learning algorithms. A large empirical literature estimates return predictability models using short rolling windows and regularized regressions such as the LASSO. In practice,

[3] Examples include OLS, weighted least squares, LASSO, ridge, elastic net, etc.

agents (or econometricians) work with limited samples, re-estimate frequently, and penalize coefficients to avoid overfitting. This empirical behaviour implies *short memory* and time–varying parameter estimates that respond strongly to recent observations.

In the next sections we build a theoretical benchmark that moves gradually from the case of infinite–memory OLS, to constant–gain learning that discounts the past, and finally to LASSO-type learning. This sequence allows us to isolate the separate roles of (i) memory (or discounting) and (ii) sparsity–inducing regularization. We show that withrecursive least squares (RLS) and decreasing gain the cross–moment consistent equilibrium (CMCE) coincides with the rational expectations equilibrium (REE) and is locally stable under learning. When agents discount the past using a constant gain, the estimated parameter oscillates around the CMCE/REE and never settles. Replacing least squares by LASSO does not eliminate these fluctuations: for small enough penalty λ we can characterize the probability that the LASSO estimate is different from zero, providing a rationale for temporary "windows of predictability" even when the underlying asset–pricing model implies unpredictable returns. We then turn to the existence and stability of CMCEs under the different learning mechanisms. [citations here]

3 Recursive Least Squares Learning

The first updating mechanism we consider is Recursive Least Squares (RLS), which is a recursive implementation of the standard Ordinary Least Squares (OLS). In OLS, the agent estimates the coefficients of a linear regression model by minimizing the sum of squared residuals. Assuming agents use OLS to estimate the parameters in the PLM (10), the OLS estimator β_t can be derived from the minimization[4] problem as

$$\beta_t = \left(\sum_{i=1}^{t} x_{i-1}^2 \right)^{-1} \sum_{i=1}^{t-1} x_{i-1} r_i. \tag{20}$$

Estimating OLS on the whole available data set is sometimes referred to as off-line estimation. Another approach is to estimate the model on-line, that is, as soon as a new data point arrives. It turns out that the estimator formula allows for a recursive formulation. Define

$$M_{t-1} = \frac{1}{t-1} \left(\sum_{i=1}^{t-1} x_{i-1}^2 \right), \quad \beta_t = \frac{1}{t-1} M_{t-1}^{-1} \sum_{i=1}^{t-1} x_{i-1} r_i. \tag{21}$$

First we establish a recursion for M_t. Start from

$$\frac{t-1}{t} M_{t-1} = \frac{1}{t} \left(\sum_{i=1}^{t-1} x_{i-1}^2 \right) \implies \frac{t-1}{t} M_{t-1} + \frac{1}{t} x_{t-1}^2 = M_t, \tag{22}$$

[4] We use the following timing: at time t the agent forecasts $r_{t+1}^e = x_t \beta_t$ which then implies a realization for r_t. Therefore, to avoid simultaneity issues, we assume that when computing the current estimate, the last available log-return observation is r_{t-1}.

that is

$$M_t = M_{t-1} + \frac{1}{t}\left(x_{t-1}^2 - M_{t-1}\right). \tag{23}$$

For β_t we have

$$\begin{aligned}\beta_t &= \beta_{t-1} + \left[(t-1)M_{t-1}\right]^{-1}\left\{x_{t-2}r_{t-1} + \left[(t-2)M_{t-2}\right]\beta_{t-1}\right\} - \beta_{t-1} \\ &= \beta_{t-1} + \left[(t-1)M_{t-1}\right]^{-1}\left\{x_{t-2}r_{t-1} - x_{t-2}^2\beta_{t-1}\right\}.\end{aligned} \tag{24}$$

The ALM associated with this learning mechanism is then

$$r_{t+1} = \log(\varepsilon_{t+1}) + \log\Big(1 - \kappa e^{w\tanh(x_t\beta_t/w)}\Big) - \log\Big(1 - \kappa e^{w\tanh(x_{t+1}\beta_{t+1}/w)}\Big). \tag{25}$$

3.1 CMCE under RLS

Proposition 1 *Under RLS there exist a unique CMCE given by $\beta = 0$. The CMCE is locally stable under learning for $\kappa \in (0, 1)$.*

Proof. In Appendix A.1.

Notice that $\kappa = \delta a^{-\gamma}\phi$ is strictly positive for all admissible values of the parameters. The upper bound implies a restriction on the combination of discount factor, risk aversion and dividend growth which is met for reasonable parameter values and also needed for the REE to be well-defined.

4 Weighted Least Squares and Constant Gain Learning

In Weighted Least Squares (WLS), the agent estimates the coefficients of a linear regression model by minimizing a weighted sum of squared residuals. The weights are typically chosen to reflect the relative importance or reliability of different observations. Assuming the linear model (10), the WLS estimator can be derived from the minimization problem

$$\min_\beta \sum_{i=1}^t w_{t,i}(y_i - \beta x_i)^2, \tag{26}$$

with solution

$$\beta = \left(\sum_{i=1}^t w_{t,i}x_i^2\right)^{-1}\sum_{i=1}^t w_{t,i}x_i y_i. \tag{27}$$

In particular, we are interested in the case where the weights decline geometrically over time[5]

$$w_{t,i} = \gamma(1-\gamma)^{t-i}, \quad \gamma \in (0, 1). \tag{28}$$

[5]In empirical applications, instead of using geometrically declining weights, it is common to use a rolling window approach, where only the most recent observations within a fixed-size window are used for estimation. However, the latter is analytically intractable. It is possible to determine a relationship between the window size and the geometric decay parameter γ such that the two methods give similar weights *overall* on the ℓ most recent observations: $\ell \approx \log(1-p)/\log(1-\gamma)$, where p is the proportion of total weight assigned to the ℓ most recent observations.

Similar to the OLS case, a recursive relationship can be derived.

Define

$$M_t = \gamma \sum_{i=1}^{t} (1-\gamma)^{t-i} x_i^2, \quad \beta_t = \gamma M_t^{-1} \sum_{i=1}^{t} (1-\gamma)^{t-i} x_i y_i. \tag{29}$$

First we establish a recursion for M_t. Start from

$$M_{t-1} = \gamma \sum_{i=1}^{t-1} (1-\gamma)^{t-1-i} x_i^2, \tag{30}$$

then

$$(1-\gamma) M_{t-1} = \gamma \sum_{i=1}^{t-1} (1-\gamma)^{t-i} x_i^2 \rightarrow (1-\gamma) M_{t-1} + \gamma x_t^2 = M_t. \tag{31}$$

That is

$$M_t = M_{t-1} + \gamma \left(x_t^2 - M_{t-1} \right). \tag{32}$$

Similarly for b_t we start from

$$\beta_{t-1} = \gamma M_{t-1}^{-1} \sum_{i=1}^{t-1} (1-\gamma)^{t-1-i} x_i y_i, \tag{33}$$

then

$$(1-\gamma)\beta_{t-1} = \gamma M_{t-1}^{-1} \sum_{i=1}^{t-1} (1-\gamma)^{t-i} x_i y_i, \tag{34}$$

and

$$\beta_{t-1}(M_t - \gamma x_t^2) = \gamma \sum_{i=1}^{t-1} (1-\gamma)^{t-i} x_i y_i \rightarrow \beta_{t-1}(1 - \gamma M_t^{-1} x_t^2) = \gamma M_t^{-1} \sum_{i=1}^{t-1} (1-\gamma)^{t-i} x_i y_i, \tag{35}$$

since

$$M_{t-1} = \frac{1}{1-\gamma}(M_t - \gamma x_t^2).$$

Adding $\gamma M_t^{-1} x_t y_t$ to both sides we obtain

$$\beta_{t-1}(1 - \gamma M_t^{-1} x_t^2) + \gamma M_t^{-1} x_t y_t = \beta_t, \tag{36}$$

which can be rearranged as

$$\beta_t = \beta_{t-1} + \gamma M_t^{-1} x_t (y_t - x_t \beta_{t-1}). \tag{37}$$

In relation to our model, this gives the same type of updating rule as RLS, with the only difference that the gain is now constant and equal to γ. Given that the gain is not decreasing and that the updating depends on the stochastic term ε_t, the learning rule cannot converge to a fixed point. However, under suitable conditions and for small values of γ, and under the same stability condition $\kappa \in (0,1)$, it can be shown that the learning dynamics will fluctuate around the CMCE.

Proposition 2 (Distribution under Constant Gain Learning) *Under Weighted Least Squares learning with geometrically declining weights with parameter γ, and for $\kappa < 1$, for small values of γ the estimated parameter β approximately follows a normal distribution*

$$\beta_t \sim \mathcal{N}\left(\beta^*, \gamma s_d^2 \left(2\sigma_x^2 \left(1 + \frac{\kappa}{1-\kappa}\right)\right)^{-1}\right), \tag{38}$$

where $\beta^* = 0$ is the cross-moment consistent equilibrium.

5 Lasso Regression

We now consider the case in which agents use LASSO regression to estimate the parameters of their PLM. The LASSO (Least Absolute Shrinkage and Selection Operator) estimator is defined as the solution to the optimization problem

$$\hat{\beta}_{\text{LASSO}} = \arg\min_{\beta} \left\{ \frac{1}{2} \left[(Y - X\beta)'(Y - X\beta) \right] + \lambda \sum_{j=1}^{k} |\beta_j| \right\},$$

where $\lambda \geq 0$ is a tuning parameter that controls the amount of regularization. The LASSO estimator can be interpreted as a trade-off between fitting the data well (minimizing the sum of squared residuals) and keeping the model simple (minimizing the absolute values of the coefficients).

In general there is no closed form solution for the LASSO estimator, since the absolute value function is not differentiable at zero. However, in the special case where X is an orthonormal matrix, the LASSO estimator can be expressed in closed form. To do so we can rewrite the optimization problem as

$$\hat{\beta}_{\text{LASSO}} = \arg\min_{\beta} \Big\{ \frac{1}{2} \left[Y'Y - 2\beta' \underbrace{X'Y}_{\beta_{\text{OLS}}} + \beta' \underbrace{X'X}_{I} \beta \right] + \lambda \sum_{j=1}^{k} |\beta_j| \Big\},$$

where the underbrackets follow from orthonormality and notice that the term $Y'Y$ does not depend on β, so it does not change the minimizer. Using $A'A = \sum_{j=1}^{k} a_j^2$, we can rewrite the problem as

$$\hat{\beta}_{\text{LASSO}} = \arg\min_{\beta} \Big\{ \sum_{j=1}^{k} \Big(-\beta_j \beta_{\text{OLS},j} + \frac{1}{2}\beta_j^2 + \lambda|\beta_j| \Big) \Big\}.$$

This problem is separable in the coefficients β_j, so we can solve for each coefficient independently. The optimization problem for each j coefficient is

$$\min_{\beta_j} \left\{ \frac{1}{2}\beta_j^2 - \beta_j \beta_{\text{OLS},j} + \lambda|\beta_j| \right\}.$$

To solve this problem, we consider two cases:

1. $\beta_{\text{OLS},j} > 0$, then β_j has to be positive, otherwise we could decrease the objective by setting $\beta_j = 0$. In this case, the problem becomes

$$\min_{\beta_j \geq 0} \left\{ \frac{1}{2}\beta_j^2 - \beta_j \beta_{\text{OLS},j} + \lambda \beta_j \right\},$$

and differentiating and setting to zero gives

$$\beta_j - \beta_{\text{OLS},j} + \lambda = 0 \Rightarrow \beta_j = \beta_{\text{OLS},j} - \lambda.$$

Imposing non-negativity yields

$$\hat{\beta}_{\text{LASSO},j} = \max\left(0, \beta_{\text{OLS},j} - \lambda\right).$$

2. $\beta_{\text{OLS},j} < 0$, then β_j has to be negative, otherwise we could decrease the objective by setting $\beta_j = 0$. In this case, the problem becomes

$$\min_{\beta_j < 0} \left\{ \frac{1}{2}\beta_j^2 - \beta_j \beta_{\text{OLS},j} + \lambda(-\beta_j) \right\},$$

and

$$\beta_j - \beta_{\text{OLS},j} - \lambda = 0 \Rightarrow \beta_j = \beta_{\text{OLS},j} + \lambda.$$

Imposing negativity yields

$$\hat{\beta}_{\text{LASSO},j} = \min\left(0, \beta_{\text{OLS},j} + \lambda\right).$$

Combining both cases, we can write the LASSO estimator for each coefficient as

$$\hat{\beta}_{\text{LASSO}} = S(\beta_{\text{OLS}}, \lambda) = \text{sign}(\beta_{\text{OLS}}) \cdot \max\left(0, |\beta_{\text{OLS}}| - \lambda\right), \tag{39}$$

where $S(\cdot)$ is the soft-thresholding operator.

In the univariate case, we othonormality is trivially satisfied by standardizing the predictor variable, so in the discussion belowe we impose the addinitional assumption

Assumption 3 *The predictor variable x_t is standardized to have zero mean and unit variance.*

Under this condition, we can still use the recursion used in recursive and constant gain learning, and then apply the soft-thresholding operator to obtain the LASSO estimate. Defining by β^O the estimate obtained from least squares learning (either RLS or WLS), and by β^L the LASSO estimate, we have The mapping between the PLM and ALM now becomes

$$\beta_{\text{OLS}} = T(\beta_{\text{LASSO}}) = -\frac{\kappa}{1-\kappa}\beta_{\text{LASSO}}, \tag{40}$$

which can be solved case by case:

1. If $\beta_{\text{OLS}} > \lambda > 0$, then $\beta_{\text{LASSO}} = \beta_{\text{OLS}} - \lambda \implies \beta_{\text{OLS}} = \lambda\kappa$.
2. If $\beta_{\text{OLS}} < -\lambda < 0$, then $\beta_{\text{LASSO}} = \beta_{\text{OLS}} + \lambda \implies \beta_{\text{OLS}} = -\lambda\kappa$.
3. If $|\beta_{\text{OLS}}|\leq \lambda$, then $\beta_{\text{LASSO}} = 0 \implies \beta_{\text{OLS}} = 0$.

Clearly, for the first two cases to be admissible we need $\lambda\kappa > \lambda \implies \kappa > 1$. Given that the dynamics are still governed by the same constant gain updating rule, we can conclude that only the third case is stable in the sense that for $\kappa < 1$ the learning dynamics will fluctuate around the fixed point. Identically to the previous case, the estimate for β_{OLS} will be distributed as a normal distribution centered around the fixed point, with variance proportional to the gain parameter γ. This implies that for small λ sometimes the estimates will be large enough that the LASSO estimate is non-zero. Given the distribution of β_{OLS}, we have:

Proposition 3 (Probability of Non-Zero LASSO Estimate) *Under Weighted Least Squares learning with geometrically declining weights with parameter γ, and for $\kappa < 1$, for small values of γ the probability that the LASSO estimate is different from zero is given by*

$$\mathbb{P}(|\beta_{OLS}|> \lambda) = 1 - \left[\Phi\left(\frac{\lambda}{\sigma_{OLS}}\right) - \Phi\left(\frac{-\lambda}{\sigma_{OLS}}\right)\right], \tag{41}$$

where $\sigma_{OLS} = \left(\gamma s_d^2 \left(2\sigma_x^2 \left(1 + \frac{\kappa}{1-\kappa}\right)\right)^{-1}\right)^{1/2}$.

6 Empirical Application

The advantage of the theoretical framework developed above is that it provides a clear guide as to how to structure an empirical investigation of return predictability under learning. We remark that when estimating our model empirically, we face the same issue that agents face in the model. We have postulated a certain model which in theoritcal section is the actual law of motion, but in reality, given we do not know the true ALM, we should regard it as the modeler's perceived law of motion. Therefore it helps to think in terms two estimations: the estimation that an agent in our model would carry out, and the estimation that we the modeler carry out. We again endow our agents with a linear PLM

$$r_{t+1} = X_t\beta_t + \eta_{t+1}, \tag{42}$$

but with respect to the theoretical counterpart, we now allow for multiple predictors collected in the vector X_t. While the closed-form results for the LASSO case, hold for orthonormal regressors only, in the empirical counterpart we are more lenient and only require that predictors are uncorrelated so that equation (39) does not hold deterministically but only approximately. We then assume that agents use a rolling window, instead of geometrically declining weights, to estimate the parameters at each point in time. Specifically at each time t the regressor matrix will contain l past observations, that is

$$X_t = \begin{bmatrix} x_{t-l} \\ x_{t-l+1} \\ \vdots \\ x_{t-1} \end{bmatrix}, \quad r_{t+1} = \begin{bmatrix} r_{t-l+1} \\ r_{t-l+2} \\ \vdots \\ r_t \end{bmatrix}. \tag{43}$$

For each rolling window, the agent then computes the corresponding LASSO estimator β_t. Given this estimator, the ALM implies the modeler's PLM

$$r_{t+1} = \log(1 - \kappa e^{X_t \beta_t}) - \log(1 - \kappa e^{X_{t+1} \beta_{t+1}}) + u_{t+1}. \tag{44}$$

This suggests that an emprical strategy to test for return predictability under learning is to estimate a LASSO regression on rolling windows of historical data, and then use the estimated coefficients to construct the model-implied returns. Finally one would estimate (44) using Non-Linear Least Squares (NLS). However as we discussed in the theoretical section, if we as researchers estimate a linear model, we would not estimate the actual parameters used by agents, but rather their trasformed counterpart implied by the T-map. This suggests that the appropriate empirical strategy is, first to estimate the (mis-specified) linear model using the exact same model we endow our agents with. Then invert the T-map to recover the parameters that agents would be using in their PLM. Finally, use these parameters to estimate the NLS model implied by the ALM.

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A Proofs

A.1 Proof of proposition 1

In a CMCE the value of β solves

$$T(\beta) - \beta = 0, \tag{45}$$

where

$$T(\beta) = \frac{\mathrm{Cov}(r_{t+1}, x_t)}{\mathrm{Var}(x_t)} = \frac{\mathrm{Cov}(r_{t+1}, x_t)}{\sigma_x^2}, \tag{46}$$

and r_{t+1} is given by (16) evaluated at the stationary distribution.

To compute $\mathrm{Cov}(r_{t+1}, x_t)$ we exploit the fact that x_t is independent of $\log(\varepsilon_{t+1})$ and x_{t+1}, which implies that

$$\mathrm{Cov}(r_{t+1}, x_t) = \mathrm{Cov}\left(g_\beta(x_t), x_t\right). \tag{47}$$

By Stein's lemma

$$\mathrm{Cov}(r_{t+1}, x_t) = \mathrm{Cov}(g_\beta(x_t), x_t) = \sigma_x^2 \mathbb{E}\left[g'_\beta(x_t)\right]. \tag{48}$$

This implies that $T(\beta) = \mathbb{E}\left[g'_\beta(x_t)\right]$.

We can show that $\mathrm{sign}\,[T(\beta)] = -\,\mathrm{sign}(\beta)$ unless $\beta = 0$. In fact, we have

$$g'_\beta(x) = -\frac{\kappa\, \beta\, e^{w \tanh(x\beta/w)}\ \mathrm{sech}^2(x\beta/w)}{1 - \kappa e^{w \tanh(x\beta/w)}}. \tag{49}$$

In the numerator, every multiplicative factor except β is strictly positive since $\kappa > 0$, $e^{w \tanh(x\beta/w)} > 0$, and $\mathrm{sech}^2(x\beta/w) > 0$. The denominator is also strictly positive since w is chosen accordingly. Hence the sign of $g'_\beta(x)$ is determined entirely by $-\beta$. In particular,

$$\mathrm{sign}\,[T(\beta)] = \mathrm{sign}\left[-\beta\, \mathbb{E}\left(\frac{\kappa\, e^{w \tanh(x\beta/w)}\ \mathrm{sech}^2(x\beta/w)}{1 - \kappa e^{w \tanh(x\beta/w)}}\right)\right] = -\,\mathrm{sign}(\beta). \tag{50}$$

Therefore $T(\beta)$ has the opposite sign of β for all $\beta \neq 0$, and the only fixed point is $\beta = 0$.

To study local stability of the CMCE under RLS learning, we use the E-stability principle of Evans and Honkapohja (2001). To do so we show how to cast the learning dynamics in the form of a Stochastic Recursive Algorithm (SRA) of the form

$$\theta_t = \theta_{t-1} + \gamma_t \mathcal{H}(\theta_{t-1}, Z_t) + \gamma_t^2 \rho_t(\theta_{t-1}, Z_t), \tag{51}$$

where θ_t is the parameter vector to be estimated, γ_t is a sequence of gains and $\mathcal{H}(\cdot)$ and $\rho_t(\cdot)$ are functions capturing the updating mechanism. Z_t is a vector of state variables that in general can be conditionally linear and evolving according to

$$Z_t = A(\theta_{t-1}) Z_{t-1} + B(\theta_t) W_t, \tag{52}$$

where W_t is a random disturbance term.

Under suitable assumptions about the gain sequence γ_t, the functional forms $\mathcal{H}(\cdot)$ and

$\rho_t(\cdot)$, and the evolution of the state variable Z_t, Evans and Honkapohja (2001) show that it is possible to associate to the SRA (51) a deterministic Ordinary Differential Equation (ODE) of the form

$$\dot{\theta} = \mathbb{E}\left[\mathcal{H}(\theta, Z)\right], \tag{53}$$

where the expectation is taken with respect to the stationary distribution of Z_t when θ is fixed. The E-stability principle states that if the ODE (53) is locally stable around a fixed point θ^*, then the SRA (51) is also locally stable around the same fixed point.

We therefore show how to apply this framework to our model.

First we can rearrange (24) to get

$$\begin{aligned} \beta_t &= \beta_{t-1} + t^{-1} M_{t-1}^{-1}\left[x_{t-2} r_{t-1} - x_{t-2}^2 \beta_{t-1}\right] \\ &\quad + t^{-2}\frac{t}{t-1} M_{t-1}^{-1}\left[x_{t-2} r_{t-1} - x_{t-2}^2 \beta_{t-2}\right]. \end{aligned} \tag{54}$$

Second, notice that r_{t-1} depends on β_{t-1} and β_{t-2} through the ALM (16). We therefore define an ancillary variable $\alpha_t = \beta_{t-1}$ so that

$$r_{t-1} = \log(\varepsilon_{t-1}) + \log\Big(1 - \kappa e^{w \tanh(x_{t-2}\alpha_{t-1}/w)}\Big) - \log\Big(1 - \kappa e^{w \tanh(x_{t-1}\beta_{t-1}/w)}\Big). \tag{55}$$

The full updating system for the parameters is then

$$\begin{aligned} \beta_t &= \beta_{t-1} + t^{-1} M_{t-1}^{-1}\left[x_{t-2} r_{t-1} - x_{t-2}^2 \beta_{t-1}\right] + t^{-2}\frac{t}{t-1} M_{t-1}^{-1}\left[x_{t-2} r_{t-1} - x_{t-2}^2 \beta_{t-2}\right] \\ \alpha_t &= \alpha_{t-1} + t^{-1}(\beta_{t-1} - \alpha_{t-1}) + t^{-2}\left(t(t-1)\right)(\beta_{t-1} - \alpha_{t-1}) \\ M_t &= M_{t-1} + \frac{1}{t}\left(x_{t-1}^2 - M_{t-1}\right), \end{aligned} \tag{56}$$

Now we appropriately define

$$\theta_t = \begin{pmatrix} \beta_t \\ \alpha_t \\ M_t \end{pmatrix}, \quad \gamma_t = t^{-1} \tag{57}$$

notice that the domain of θ_t is $\mathbb{R} \times \mathbb{R} \times \mathbb{R}_+$, since M_t is a sum of squared variables and therefore strictly positive. We also define

$$\underbrace{\begin{pmatrix} x_{t-1} \\ x_{t-2} \\ \log(\varepsilon_{t-1}) \end{pmatrix}}_{Z_t} = \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}}_{A} \underbrace{\begin{pmatrix} x_{t-2} \\ x_{t-3} \\ \log(\varepsilon_{t-2}) \end{pmatrix}}_{Z_{t-1}} + \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}}_{B} \underbrace{\begin{pmatrix} x_{t-1} \\ \log(\varepsilon_{t-1}) \end{pmatrix}}_{W_t}. \tag{58}$$

The state evolution clearly satisfies assumpotions (B.1) and (B.2) in Section 6.2.1 of Evans and Honkapohja (2001), since both x_t and $\log(\varepsilon_t)$ are normal i.i.d variables with finite moments of all orders. Assumption (A.1) is satisfied by construction of the gain sequence $\gamma_t = t^{-1}$. For assumotion (A.2) and (A.3) write

$$\mathcal{H}(\theta_{t-1}, X_t) = \begin{pmatrix} M_{t-1}^{-1}\left[x_{t-2}r_{t-1} - x_{t-2}^2 b_{t-1}\right] \\ b_{t-1} - a_{t-1} \\ x_{t-1}^2 - M_{t-1} \end{pmatrix}, \tag{59}$$

$$\rho_t(\theta_{t-1}, X_t) = \begin{pmatrix} \frac{t}{t-1} M_{t-1}^{-1}\left[x_{t-2}r_{t-1} - x_{t-2}^2 b_{t-2}\right] \\ (t(t-1))(b_{t-1} - a_{t-1}) \\ 0 \end{pmatrix}. \tag{60}$$

To verify Assumption (A.2), we observe that each component of $\mathcal{H}(\theta_{t-1}, Z_t)$ is continuous in θ on any compact subset of the parameter space. The transformation $w\tanh(x\theta/w)$ is smooth and bounded, which implies that

$$\log\Big(1 - \kappa e^{\,w\tanh(x\theta/w)}\Big)$$

remains uniformly bounded for all (θ, x) whenever $\kappa e^w < 1$. Consequently, the return innovation r_{t-1} is bounded by a constant plus $|\log \varepsilon_{t-1}|$, and the expressions $x_{t-2}r_{t-1}$, $x_{t-2}^2 b_{t-1}$, and x_{t-1}^2 grow at most polynomially in $|x_t|$. Since the exogenous process (x_t) is strictly stationary and ergodic with $\mathbb{E}[x_t^2] < \infty$ and $\mathbb{E}[x_t^4] < \infty$, all expectations defining $h(\theta) = \mathbb{E}[\mathcal{H}(\theta, X_t)]$ exist and $\mathcal{H}$ satisfies the polynomial-growth and continuity requirements of Assumption (A.2).

Assumption (A.3) holds provided that $\mathcal{H}(\theta, Z)$ is twice continuously differentiable in θ with second derivatives bounded on every compact set Q. In our case, the mappings

$$\theta \mapsto w\tanh(x\theta/w), \qquad \theta \mapsto \log\Big(1 - \kappa e^{\,w\tanh(x\theta/w)}\Big)$$

are C^∞ with all derivatives bounded uniformly on compacts, because $\tanh(\cdot)$ and its derivatives are bounded and the restriction $\kappa e^w < 1$ keeps the logarithmic term uniformly away from its singularity. It follows that each component of $\mathcal{H}$ is twice continuously differentiable in θ with bounded second derivatives on compact sets, which verifies Assumption (A.3).

The E-stability principle then applies and we can study the stability of the equilibrium by analyzing the associated ODE. We first obtain

$$h(\theta) = \lim_{t\to\infty} \mathbb{E}\left[\mathcal{H}(\theta, \bar{Z}_t)\right] = \begin{pmatrix} M^{-1}\left[\mathbb{E}(x_{t-2}r_{t-1}) - \beta\sigma_x^2\right] \\ \beta - \alpha \\ \sigma_x^2 - M \end{pmatrix} = \begin{pmatrix} M^{-1}\left[(T(\alpha) - \beta)\sigma_x^2\right] \\ \beta - \alpha \\ \sigma_x^2 - M \end{pmatrix} \tag{61}$$

where $\bar{Z}_t$ refers to the evolution of the state variable, under fixed values of the parameters.

Then we evaluate the stability of the cross-moment consistent equilibrium by studying the associated ordinary differential equation (ODE)

$$\frac{d\theta}{d\tau} = h(\theta). \tag{62}$$

Therefore we compute the Jacobian matrix of $h(\theta)$ evaluated at the equilibrium point $\theta = (0, 0, \sigma_x^2)$

$$J = \begin{pmatrix} -1 & T'(0) & 0 \\ 1 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{63}$$

The eigenvalues of J are given by

$$\lambda_{1,2} = -1 \pm i\sqrt{\frac{\kappa}{1-\kappa}}, \quad \lambda_3 = -1, \tag{64}$$

since

$$T'(\alpha) = -\mathbb{E}\left[\frac{\kappa x^2 e^{w\tanh(x\alpha/w)}\text{sech}^2(x\alpha/w)}{1-\kappa e^{w\tanh(x\alpha/w)}}\right]\left(\sigma_x^2\right)^{-1}. \tag{65}$$

therefore the ODE is asymptotically stable for the whole range of $\kappa \in (0, 1)$.